

Labwork 11: Bitwise Operations

General Rules

- Use 32-bit unsigned integers.
 - Bits are numbered from the right starting at 0 (LSB). So bit 0 is the least significant bit.
 - When counting from the left (MSB side), bit positions are 31, 30, 29, ... down to 0.
 - Unless the question says "modify in-place", return the new value.
 - Do not use loops or arrays unless the question explicitly requires them.
 - All return values for "get" questions must be exactly 0 or 1.
 - Use bitwise operators (&, |, ^, ~, <<, >>). Do not use division or modulo.
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Questions

Q1. Get a bit (right side)

Write a function that returns the value of the 5th bit from the right of a given unsigned integer.

- Input: an unsigned integer `x`
- Output: 0 if the 5th bit is 0, 1 if the 5th bit is 1
- Clarify: 5th bit from the right means bit position 4
- Requirement: do not change any other bits of `x`

Q2. Set a bit (right side)

Write a function that sets the 3rd bit from the right to 1 and returns the new integer.

- Input: an unsigned integer `x`
- Output: a new unsigned integer with the 3rd bit forced to 1
- Clarify: 3rd bit from the right means bit position 2
- Requirement: all other bits must remain the same as in `x`

Q3. Clear a bit (right side)

Write a function that clears the 6th bit from the right to 0 and returns the new integer.

- Input: an unsigned integer `x`
- Output: a new unsigned integer with the 6th bit forced to 0
- Clarify: 6th bit from the right means bit position 5
- Requirement: all other bits must remain the same as in `x`

Q4. Toggle a bit (right side)

Write a function that toggles the 1st bit from the right and returns the new integer.

- Input: an unsigned integer `x`
- Output: a new unsigned integer where the 1st bit from the right is flipped
- Clarify: 1st bit from the right means bit position 0

- Requirement: only the target bit may change

Q5. Replace a bit (right side)

Write a function that sets the 4th bit from the right to a given value **b**.

- Input: an unsigned integer **x** and an unsigned integer **b** where **b** is 0 or 1
- Output: a new unsigned integer with the 4th bit set to **b**
- Clarify: 4th bit from the right means bit position 3
- Requirement: if **b** is 0, the bit must be cleared; if **b** is 1, the bit must be set

Q6. Get a bit (left side)

Write a function that returns the value of the 3rd bit from the left (MSB side).

- Input: an unsigned integer **x**
- Output: 0 or 1 based on the 3rd bit from the left
- Clarify: 3rd bit from the left means bit position 29
- Requirement: return exactly 0 or 1

Q7. Clear a bit (left side)

Write a function that clears the 5th bit from the left to 0 and returns the new integer.

- Input: an unsigned integer **x**
- Output: a new unsigned integer with the 5th bit from the left forced to 0
- Clarify: 5th bit from the left means bit position 27
- Requirement: all other bits must remain the same as in **x**

Q8. Set a bit (left side)

Write a function that sets the 2nd bit from the left to 1 and returns the new integer.

- Input: an unsigned integer **x**
- Output: a new unsigned integer with the 2nd bit from the left forced to 1
- Clarify: 2nd bit from the left means bit position 30
- Requirement: all other bits must remain the same as in **x**

Q9. Get a bit (right side)

Write a function that returns the value of the 2nd bit from the right of a given unsigned integer.

- Input: an unsigned integer **x**
- Output: 0 if the 2nd bit is 0, 1 if the 2nd bit is 1
- Clarify: 2nd bit from the right means bit position 1
- Requirement: return exactly 0 or 1